

REVIEW EXERCISE 1

1. **Write the following numbers in words:**
(i) 246,416,279 (ii) 905,407,608
2. **Write the following numbers in numerals:**
(i) Seventy five million, twenty six thousand four hundred twenty.
(ii) Four hundred five million, seven hundred forty five thousand eight hundred six.
3. **Add the following:**
(i) 205, 617,291 and 5,412,306
(ii) 4,000,405 and 20136,999
(iii) 214,308,196 and 523,410,018
4. **Subtract the following:**
(i) 412,326,917 from 624,005,123
(ii) 96,105,892 from 712,342,445
(iii) 234,596,501 from 641,884,962
5. **Solve:**
(i) $1,243 \times 10$ (ii) $962,345 \times 45$ (iii) $56,729 \times 40$
(iv) $612,378 \times 962$ (v) $405,617 \times 1000$
6. **Divide:**
(i) 753400 by 30 (ii) 269817 by 356
7. **Solve:**
(i) $180 \div 10 \times (50 \text{ of } 2 \div 4)$ (ii) $48 - (5 \text{ of } 2 + 12)$

8. Verify:

(i) $2 \times (4 + 7) = (2 \times 4) + (2 \times 7)$

(ii) $5 \times (96 - 34) = (5 \times 96) - (5 \times 34)$

9. A water tanker holds **24,541** litres of water. How much will be in **35** such tankers?

10. A box contains **30** chalks. How many such boxes are required to pack **24,660** chalks?

HCF AND LCM

(Highest Common Factor and Least Common Multiple)

2.1 HCF

2.1.1 Find HCF of three numbers, up to 2-digits

In the previous class, we learned about HCF by using Venn diagram and prime factorization method, LCM by using common multiples and prime factorization. Now we will learn HCF and LCM in more detail.

HCF means Highest Common Factor. It is of two or more natural numbers. It is the greatest common factor (divisor) of the given natural numbers. HCF is also called Greatest Common Divisor (GCD).

Methods for finding the HCF or GCD are given below.

(1) **Prime Factorization method**

(2) **Division method**

1. **Prime Factorization method****Procedure:**

- Write prime factors of all the given numbers.
- Find common factors from all the prime factors.
- Write the product of all common factors.
- The product of common factors is the required HCF.

Consider the following examples:

Example 1.

Find HCF of **40** and **50** by using prime factors method.

Solution:

Prime factors of 40 = $2 \times 2 \times 2 \times 5$

Prime factors of 50 = $2 \times 5 \times 5$

Common prime factors are 2 and 5

Product of common factors = $2 \times 5 = 10$

So, HCF of **40** and **50** is **10**.

2	40	2	50
2	20	5	25
2	10	5	5
5	5		1
	1		

Example 2. Find HCF of 18, 30 and 36.

Solution:

Prime factors of 18 = $2 \times 3 \times 3$

Prime factors of 30 = $2 \times 3 \times 5$

Prime factors of 36 = $2 \times 2 \times 3 \times 3$

Common factors of all the prime factors are 2 and 3.

Product of common factors = $2 \times 3 = 6$

So, HCF of 18, 30 and 36 is 6.

2	18
3	9
3	3
	1

2	30
3	15
5	5
	1

2	36
2	18
3	9
3	3
	1

2. By Division Method

We have learnt to find the HCF by prime factors method. Now we will learn to find HCF of two or more numbers by division method.

Let us consider the following examples.

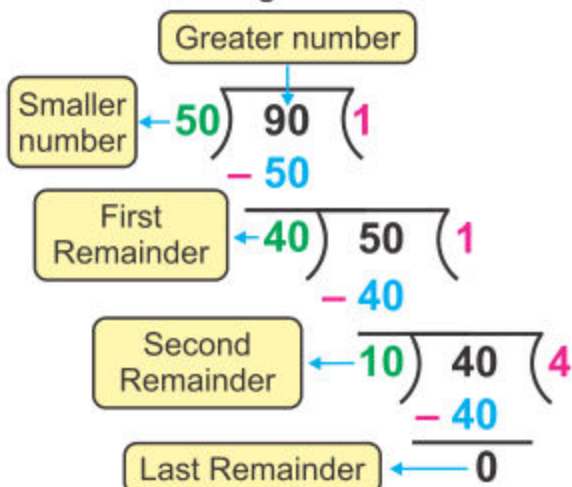
Example 1: Find HCF of 50 and 90 by division method.

Solution:

Given two numbers are 50 and 90.

Larger number is 90 and smaller number is 50.

First we divide greater number 90 by smaller number 50.



Step 1: $90 \div 50 = 1$, remainder 40

First remainder is 40.

Step 2: Again we divide 50 by 40.

We get $50 \div 40 = 1$, remainder 10

Step 3: Lastly $40 \div 10 = 4$, remainder 0

Hence, HCF of 50 and 90 = 10

Teacher's Note

Teacher should revise the methods of finding HCF learnt in previous class. Teacher should also help to understand the procedure of prime factorization method.

Example 2. Find HCF of 30, 50 and 80.

Solution:

First find HCF of any two given numbers; say 30 and 50.

(i) Divide greater number 50 by smaller number 30.

Greater number ↓	$\begin{array}{r} 30 \overline{) 50} (1 \\ - 30 \\ \hline 20 \end{array}$	So, $50 \div 30 = 1$, remainder 20
Smaller number ←	$\begin{array}{r} 20 \overline{) 30} (1 \\ - 20 \\ \hline 10 \end{array}$	Again $30 \div 20 = 1$, remainder 10
First Remainder ←	$\begin{array}{r} 10 \overline{) 20} (2 \\ - 20 \\ \hline 0 \end{array}$	Lastly $20 \div 10 = 2$, remainder 0
Second Remainder ←	0 →	Last Remainder

Hence, HCF of 30 and 50 = 10

Now we have to find HCF of 10 and the remaining number 80 by division method.

Divide the greater number 80 by smaller number 10.

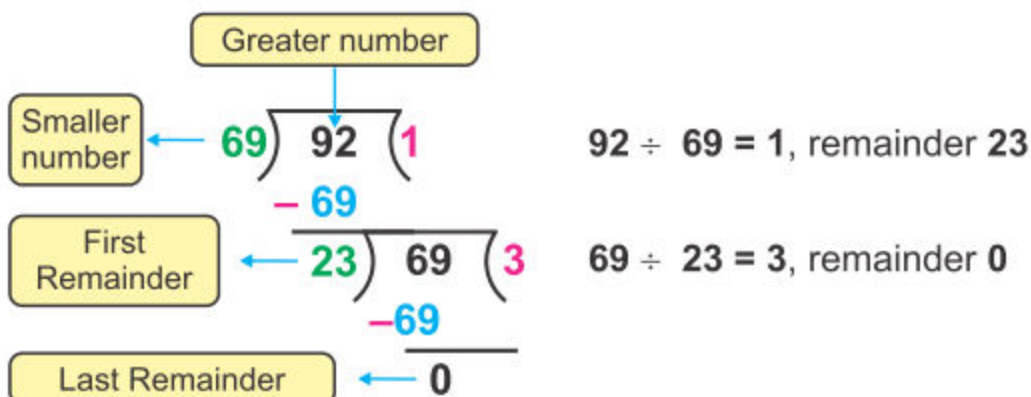
$\begin{array}{r} 10 \overline{) 80} (8 \\ - 80 \\ \hline 0 \end{array}$	80 is exactly divided by 10. $80 \div 10 = 8$, remainder 0. So, 10 is HCF of 10 and 80.
Last Remainder	

Hence, HCF of 30, 50 and 80 = 10

Example 3. Find HCF of 46, 69 and 92.

Solution:

- (i) Find HCF of any two given numbers, say 69 and 92.
- (ii) Divide greater number 92 by smaller number 69.



So, HCF of 69 and 92 = 23

Again find HCF of 23 and the third number 46 by division method.

Divide the greater number 46 by smaller number 23.

$$\begin{array}{r} 46 \overline{) 23} \quad (2) \\ \underline{-46} \\ 0 \end{array}$$

46 is exactly divided by 23
 $46 \div 23 = 2$, remainder 0.
 So, HCF of 46 and 23 = 23

Hence, 23 is HCF of 46, 69 and 92.

EXERCISE 2.1

A. Find HCF of the following by Prime Factorization method.

- | | | |
|--------------------|--------------------|--------------------|
| (1) 16 and 28 | (2) 27 and 36 | (3) 24 and 56 |
| (4) 28 and 42 | (5) 44 and 66 | (6) 52 and 78 |
| (7) 20, 60 and 80 | (8) 32, 48 and 96 | (9) 35, 49 and 63 |
| (10) 26, 39 and 65 | (11) 45, 75 and 90 | (12) 21, 35 and 63 |

B. Find HCF of the following by using division method.

- | | | |
|--------------------|--------------------|--------------------|
| (1) 28 and 70 | (2) 66 and 88 | (3) 57 and 95 |
| (4) 51 and 85 | (5) 48 and 80 | (6) 54 and 90 |
| (7) 40, 60 and 80 | (8) 36, 60 and 96 | (9) 32, 48 and 80 |
| (10) 60, 75 and 90 | (11) 42, 70 and 84 | (12) 63, 72 and 81 |

2.2 LCM

2.2.1 Find LCM of four numbers up to 2-digits.

LCM means Least Common Multiple. Least Common Multiple is of two or more natural numbers.

LCM is the smallest natural number which is a multiple of the given numbers.

There are two methods to find the LCM of two or more numbers.

- (a) Prime factorization method
- (b) Division method

(a) LCM by using prime factorization method.

Steps:

- Write all prime factors of each number.
- Write all common and non-common prime factors out of these prime factors.
- The product of all common and uncommon prime factors is the required LCM of the given numbers.

Example 1.

Find LCM of 32 and 40 by Prime Factorization method.

Solution: Given two numbers are 32 and 40

Prime factorization of 32 = $2 \times 2 \times 2 \times 2 \times 2$

Prime factorization of 40 = $2 \times 2 \times 2 \times 5$

Common factors of both prime factorizations are 2, 2, 2

Non-common factors of both prime factorizations are 2, 2, 5.

Product of common factors = $2 \times 2 \times 2 = 8$

Product of non-common factors = $2 \times 2 \times 5 = 20$

LCM is product of common and non-common factors
= $8 \times 20 = 160$

So, the required LCM is 160.

2	32
2	16
2	8
2	4
2	2
	1
2	40
2	20
2	10
5	5
	1

Teacher's Note

Teacher should revise the methods of LCM with concept. And help the students to understand the procedure of prime factorization of four numbers.

Example 2.

Find LCM of 18, 24, 36 and 60.

Solution:Prime factors of 18 = $2 \times 3 \times 3$ Prime factors of 24 = $2 \times 2 \times 2 \times 3$ Prime factors of 36 = $2 \times 2 \times 3 \times 3$ Prime factors of 60 = $2 \times 2 \times 3 \times 5$

Common prime factors are 2, 3

Common prime factors of in 2 or 3 numbers and uncommon prime factors are 2, 2, 3, 5

LCM is product of all these factors

$$= (2 \times 3) \times (2 \times 2 \times 3 \times 5) = (6) \times (60) = 360$$

So, the required LCM is **360**.

2	18
3	9
3	3
	1

2	24
2	12
2	6
3	3
	1

2	36	2	60
2	18	2	30
3	9	3	15
3	3	5	5
	1		1

(b) LCM by using division method

We have learnt to find the LCM by prime factorization method. Now we will learn to find LCM of four numbers by division method. Let us consider the following examples.

Example 1: Find LCM of 16 and 20 by division method.**Solution:**

$$\text{LCM} = 2 \times 2 \times 2 \times 2 \times 5 = 80$$

Therefore LCM of 16 and 20 = 80

Steps:

- Write the numbers as shown above.
- Divide both numbers by the smallest prime number which exactly divides at least one of the number.
- Write the quotient in each case below the number.
- If a number cannot be divided exactly, write the number as it is in the next row.
- Keep on dividing until the last row has only 1 under each number.
- Product of all divisors gives the LCM.

2	16, 20
2	8, 10
2	4, 5
2	2, 5
5	1, 5
	1, 1

Example 2. Find the LCM of 12, 36 and 60 by division method.

Solution:

2	12, 36, 60
2	6, 18, 30
3	3, 9, 15
3	1, 3, 5
5	1, 1, 5
	1, 1, 1

Thus, $\text{LCM} = 2 \times 2 \times 3 \times 3 \times 5 = 180$

Example 3: Find LCM of 50, 60, 75 and 90 by division method.

Solution:

2	50, 60, 75, 90
2	25, 30, 75, 45
3	25, 15, 75, 45
3	25, 5, 25, 15
5	25, 5, 25, 5
5	5, 1, 5, 1
	1, 1, 1, 1

$\text{LCM} = 2 \times 2 \times 3 \times 3 \times 5 \times 5 = 900$

Hence the required LCM of the given numbers is 900.

EXERCISE 2.2

A. Find LCM of the following by Prime factorization method.

- (1) 36 and 54 (2) 33 and 55 (3) 52 and 78
 (4) 16, 24 and 40 (5) 27, 48 and 72 (6) 50, 80 and 90
 (7) 56, 84 and 98 (8) 44, 66 and 99 (9) 25, 50 and 75
 (10) 15, 25, 30 and 45 (11) 10, 20, 32 and 40 (12) 12, 24, 48 and 54

B. Find LCM of the following by division method.

- (1) 16 and 24 (2) 20 and 25 (3) 36 and 48
 (4) 27, 36 and 45 (5) 28, 35 and 63 (6) 48, 64 and 96
 (7) 54, 72 and 90 (8) 55, 88 and 99 (9) 60, 70 and 80
 (10) 8, 12, 32 and 48 (11) 18, 27, 36 and 45 (12) 20, 40, 60 and 80

2.2.1 Solve the real life problems involving HCF and LCM.

To solve real life problems with HCF or LCM it is important to read the question carefully and identify what needs to be found out.

Let us consider the following examples:

Example 1. Find the least number of toffees which can be equally distributed among 15, 30 or 60 friends.

Solution: For equal distribution of least number of toffees, we have to find the LCM of 15, 30 or 60.

2	15, 30, 60
2	15, 15, 30
3	15, 15, 15
5	5, 5, 5
	1, 1, 1

So, $LCM = 3 \times 5 \times 2 \times 2 = 60$

Thus, the required number of toffees are **60**.

Example 2: Four containers have 32, 40, 56 and 72 litres capacity. Find the capacity of the largest utensil by which these containers can be filled exactly.

Solution:

2	32, 40, 56, 72
2	16, 20, 28, 36
2	8, 10, 14, 18
	4, 5, 7, 9

Highest Common Factor (HCF) = $2 \times 2 \times 2 = 8$

Therefore each container will be filled **8** litres.

EXERCISE 2.3

1. The lowest number, which is divisible by **15, 25, 40** and **75**.
2. Find the greatest number that will divide **42, 66** and **78**.